School of Computing Science and Artificial Intelligence

DSN2098 - Project Exhibition – I

Smart Medical Record Management System

A Structured Framework for Data Quality, Integrity, and Privacy in Electronic Health Records (MedVault)

Domain: Web Development and Cloud in Health Informatics

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I. Project Overview

1.1 Abstract

Modern Electronic Health Record (EHR) infrastructures are essential for clinical workflows, epidemiology, and automated clinical triage. However, operational hospital environments suffer from three fundamental vulnerabilities: (1) systemic data quality degradation from transcription errors, out-of-range sensor readings, and incomplete profiles; (2) vulnerability to covert database tampering and insider attacks lacking immutable non-repudiation; and (3) severe patient privacy risks through membership inference attacks and unauthorized record exfiltration.

MedVault is our comprehensive attempt to resolve this triad through a unified, multi-tiered software architecture. The system integrates: (i) an algorithmic Data Quality Assurance (DQA) pipeline featuring a formal Physiological Plausibility Metric (PPM) that enforces univariate survival boundaries and bivariate conservation laws (Wintrobe invariants: HCT $\approx 3 \times$ HGB); (ii) a high-throughput tamper-evident cryptographic ledger using balanced binary Merkle trees and Keyed-Hash Message Authentication Code (HMAC-SHA256) epoch chaining to guarantee non-repudiation and sub-millisecond audit proofs; and (iii) a standardized clinical triage classification engine protected by bounded (ε, δ)-Differential Privacy using the calibrated Laplace mechanism, combined with deterministic emergency red-flag overrides for acute hematological conditions (e.g., severe thrombocytopenia).

The framework was empirically benchmarked on a real-world multi-center clinical dataset of **5,074 patient complete blood count (CBC) records** (2,046 In-Patient admissions, 3,028 Out-Patient routine cases) across ten physiological biomarkers. MedVault achieves a triage classification accuracy of **85.8%** and an Area Under the ROC Curve (AUC-ROC) of **0.892**, outperforming standard Random Forest (85.1%), Multilayer Perceptron (84.7%), and Support Vector Machines (83.4%). Under calibrated differential privacy ($\varepsilon = 1.0$), the model retains 84.6% accuracy while reducing Membership Inference Attack vulnerability from 78.6% to 53.8% (near random chance). Cryptographic audit verification incurs only 0.13 ms per transaction (>2,500 TPS), confirming that MedVault delivers enterprise-grade data quality, auditability, and privacy without introducing latency bottlenecks into mission-critical clinical environments.

1.2 Acknowledgement

First and foremost, we would like to thank the Lord Almighty for His presence, wisdom, and immense blessings throughout our project work.

We wish to express our heartfelt gratitude to the **Head of the Department** and the faculty leadership of the School of Computing Science Engineering and Artificial Intelligence (SCAI), VIT Bhopal University, for their continuous support, encouragement, and visionary guidance in facilitating this academic inquiry.

We would like to express our deep appreciation to our **Faculty Project Guide**, for continually mentoring, reviewing our architectural designs, and actively participating in our progress, giving critical insights that elevated this work from a conceptual prototype to a production-grade platform.

We extend our sincere thanks to all the technical, lab, and teaching staff of VIT Bhopal University who supported us throughout development, as well as our parents and families whose unwavering patience and encouragement made this project possible.

II. Introduction

2.1 Background and Motivation

Healthcare delivery across the globe is experiencing an unprecedented transformation driven by digital health records, decentralized clinical laboratories, and artificial intelligence. Digital record-keeping promises to streamline patient handoffs, reduce medical errors, and provide continuous longitudinal histories. The World Health Organization (WHO) and international medical bodies have underscored that timely detection of physiological anomalies and structured data sharing are pivotal to reducing preventable inpatient mortality.

However, clinical monitoring in operational hospitals requires interpreting complex, heterogeneous laboratory signals. Complete Blood Count (CBC) panels are among the most frequently ordered diagnostic investigations in acute medicine. Clinicians must simultaneously assess hematocrit (packed red cell volume), hemoglobin levels, white blood cell counts, and platelet counts to determine whether a patient requires immediate hospital admission (In-Patient Care) or can be safely managed with routine outpatient follow-up (Out-Patient Care).

Existing clinical systems are severely fragmented: hospital information systems record laboratory numbers without validating physiological conservation laws; traditional relational databases are vulnerable to silent tampering by insiders; and predictive triage algorithms deployed without privacy defenses risk leaking confidential patient profiles. This motivated the development of **MedVault**: an integrated, auditable, and privacy-preserving platform that converts clinical blood counts into trustworthy, actionable triage intelligence.

2.2 Problem Statement

Single-sentence problem statement:

> *There is no accessible, integrated, and verifiable healthcare data management system that simultaneously enforces biological data quality, tamper-evident cryptographic auditability, and provable differential privacy for acute clinical triage in operational environments.*

Detailed problem points:

- **Data Quality Degradation:** Upstream transcription errors, sensor drift, and missing biomarker values propagate into clinical workflows, inducing algorithmic hallucination and catastrophic misdiagnoses.
- **Covert Database Tampering:** Relational database systems (RDBMS) administered centrally allow privileged database administrators (DBAs) or intruders to silently alter diagnostic history without leaving immutable cryptographic proof.
- **High Blockchain Overhead:** Proposed blockchain solutions (e.g. Ethereum, Hyperledger) suffer from crippling latency (15–300 TPS), high gas fees, and violate GDPR Article 17 ('Right to Erasure').
- **Inferential Privacy Leaks:** Clinical triage calculators leak sensitive patient attributes through membership inference attacks (MIA) when queried by untrusted third-party researchers.
- **Operational Fragility:** Prototype healthcare ML systems collapse when model artifacts are missing or inputs fall outside expected statistical ranges.

What must be solved: Provide a resilient, evidence-linked, dual-serving solution that (1) validates physiological plausibility before storage, (2) builds lightweight Merkle hash chains for tamper-evidence, (3) executes (ε, δ)-differentially private triage, and (4) runs live on modest cloud infrastructure with graceful fallback behavior.

2.3 Objectives of the Project

Primary Objectives (Core Deliverables):

1. **Develop a Three-Tier Architecture:** Decouple clinical data quality assurance, cryptographic ledger sealing, and AI risk prediction into an integrated pipeline.

2. **Build a Clinical Data Quality Assurance Engine:** Implement the Physiological Plausibility Metric (PPM) to reject contaminated records using biological conservation laws (Wintrobe invariants: HCT ≈ 3 × HGB).

3. **Construct a High-Throughput Cryptographic Ledger:** Implement binary Merkle trees and HMAC-SHA256 epoch chaining delivering O(log N) verification in < 1.84 ms.

4. **Deliver an Interactive Fullstack Web Portal:** Build a responsive frontend in React 18, Vite, and Tailwind CSS v4, connected to an Express.js backend and deployed live on Render.

Secondary Objectives (Value-Add Capabilities):

5. **Explainability & Clinician Guidance:** Provide human-readable medical reasoning, risk factor breakdowns, and actionable doctor recommendations for each triage decision.

6. **Emergency Red-Flag Traps:** Deterministic rule-based triggers for life-threatening emergencies (e.g. Platelets < 50k, WBC > 14k, Hemoglobin < 8 g/dL).

7. **Differential Privacy Bounds:** Perturb risk scores via calibrated Laplace noise ($\varepsilon = 1.0$) to suppress membership inference vulnerability to ~53%.

8. **Resilient Mock Fallbacks:** Implement rule-based heuristics that ensure continuous demoability even if serialized model files are missing.

Quantitative / Success Criteria:

- Clinical triage classification accuracy ≥ 85.0% and AUC-ROC ≥ 0.88 on the 5,074 patient cohort.
- Cryptographic ingestion and verification latency < 2.0 ms per record (> 2,500 TPS).
- Functional fullstack web portal live on cloud infrastructure accessible via public HTTPS URL.

2.4 Scope of the Work

In-scope (what MedVault implements in this project):

- Ingestion, validation, and analysis of 5,074 clinical complete blood count patient records.
- Training and evaluation of baseline and ensemble models: Logistic Regression, Random Forest, SVM, and MLP.
- Interactive web application with Home Portal, Live Patient Records Explorer, and AI Clinical Diagnostic Analyzer Modal.
- Automated cloud deployment blueprint (`render.yaml`, `Dockerfile`, `Procfile`) on Render Web Services.
- Unit testing, boundary testing, synthetic corruption resilience testing, and benchmark logging.

Out of scope (explicit exclusions for this phase):

- Direct physical hardware interfacing with bedside hematology analyzers (uses standard JSON/FHIR batch feeds).
- Clinical certification as an FDA/CE-marked diagnostic medical device (educational and research prototype).
- Real-time hospital paging hardware integration (contact and appointment booking APIs are software-backed).

Assumptions & Constraints: Input parameters conform to standardized hematological units (% for HCT, g/dL for HGB, k/µL for platelets/WBC). Models are evaluated on curated clinical data; prospective validation in live hospital ICUs is planned for future work.

Ethical Note and Usage Disclaimer: MedVault is an assistive educational and research system. All outputs include a clinical disclaimer stating that algorithmic triage does not replace professional physician judgment.

III. Literature Review

3.1 Clinical Guidelines in Laboratory Hematology

Clinical laboratory hematology is governed by strict international quality standards established by the Clinical and Laboratory Standards Institute (CLSI) and the World Health Organization (WHO). Complete blood counts represent the first-line screening for infection, hematologic malignancy, anemia, and bleeding diatheses. A critical principle in laboratory medicine is *biological plausibility*: healthy and pathological human blood follows rigorous physiological bounds. Specifically, Wintrobe's classic indices link packed red cell volume (hematocrit) and hemoglobin concentration (HCT $\approx$ 3 $\times$ HGB). When laboratory analyzers experience dilution errors or transcription errors, this ratio fractures. MedVault operationalizes this clinical literature by embedding Wintrobe invariants directly into automated DQA.

3.2 Existing ML Models in Healthcare Prediction

Machine learning models have achieved remarkable success in predicting clinical risk from structured tabular EHR data. Baseline classifiers such as Logistic Regression and K-Nearest Neighbors (KNN) offer high interpretability but struggle with non-linear clinical boundaries. Ensemble techniques—notably Random Forests (Breiman, 2001) and Gradient Boosted Decision Trees (Friedman, 2001)—consistently outperform traditional methods on tabular clinical datasets by handling correlated features and missing values. Recent literature published in *Lancet Digital Health* and *Nature Medicine* highlights that ensemble models trained on hematological markers can predict acute inpatient admission with high sensitivity, provided robust preprocessing and outlier handling are enforced.

3.3 Electronic Health Record Integrity & Tamper-Evident Logging

Ensuring data integrity in digital health systems has historically relied on database access logs and relational audit tables. However, Crosby and Wallach (2009) demonstrated that mutable database logs offer zero protection against privileged insiders who can rewrite historical records. While blockchain-based healthcare frameworks (e.g., FHIRChain by Zhang et al., 2018) proposed smart contracts for access provenance, empirical studies show that blockchain consensus mechanisms introduce crippling latency and excessive transaction fees. Merkle trees (Merkle, 1987) combined with cryptographic hash chaining provide a proven mathematical alternative: they guarantee forward-secure tamper-evidence and logarithmic verification proofs without consensus overhead.

3.4 Limitations of Current Approaches

Despite technological progress, current clinical record systems exhibit four critical flaws:

1. **Siloed Architecture:** Quality verification, cryptographic storage, and predictive inference are developed as disconnected tools.
2. **Static Schema Checks:** SQL `CHECK` constraints verify only column types, ignoring multivariate biological consistency.
3. **Black-Box Predictions:** Commercial triage calculators output opaque risk percentages without medical reasoning or clinical citations.
4. **Unbounded Privacy Vulnerability:** Open APIs exposing trained models are vulnerable to Membership Inference Attacks (MIA), revealing whether an individual patient's record was used in training.

IV. Existing Work and Limitations

4.1 Prior Research Tools

Several academic and commercial systems have explored digital health records and predictive triage. Hospital EHR vendors (Epic, Cerner) provide centralized clinical document repositories with role-based access control (RBAC). In academic literature, early risk stratification systems (e.g., APACHE-II, SOFA scoring) utilized rule-based point systems based on vital signs. More recently, deep learning EHR frameworks (Rajkomar et al., 2018) achieved strong predictive accuracy across hospital readmissions. However, these systems remain tightly coupled to monolithic enterprise hospital infrastructure, making them inaccessible to smaller clinics and research testbeds.

4.2 Gaps Identified

Through empirical review of existing tools, five critical gaps were identified:

• **Absence of Automated Physiological Plausibility Filtering:** Erroneous entries (e.g. negative platelet counts, impossible hemoglobin levels) are committed into ledgers.

• **Vulnerability to Retrospective DBA Modification:** Internal database admins can alter historical lab results without invalidating an immutable proof.

• **High Blockchain Friction:** Existing decentralized health networks suffer from high latency (>15 seconds per block) and high gas costs.

• **Lack of Differential Privacy in Triage:** Model endpoints leak individual training record membership when queried repeatedly.

• **Operational Fragility in Demos:** Systems crash if pickled model files are corrupt or offline.

4.3 How MedVault Addresses These Gaps

MedVault directly bridges these gaps through four architectural innovations:

1. **Physiological Plausibility Metric (PPM):** An algorithmic filter that evaluates multivariate invariants before persistence.

2. **Lightweight Cryptographic Ledger:** In-memory binary Merkle trees and HMAC forward-secure chaining delivering >2,500 TPS with sub-2ms verification.

3. **Calibrated Differentially Private Triage:** Laplace noise injection bounded to sensitivity $\Delta f / \varepsilon$, suppressing membership inference to near random chance.

4. **Unified Dual-Serving Cloud Deployment:** Express.js backend serving both React frontend assets and REST APIs on a single port with zero CORS friction.

V. Software and Hardware Requirements

5.1 Hardware Requirements

MedVault is engineered to be lightweight, eliminating the need for expensive multi-GPU clusters or specialized server hardware:

• **Processor:** 2.0 GHz Dual-Core x86-64 / Apple Silicon (M1/M2/M3) or higher.
• **RAM:** Minimum 4 GB system memory (8 GB recommended for simultaneous fullstack dev).
• **Storage:** Minimum 500 MB free disk space for Node.js modules, patient dataset, and build artifacts.
• **Display:** 1280 × 720 minimum screen resolution (responsive on mobile, tablet, and desktop).
• **Connectivity:** Standard internet access for cloud deployment and API communication.

5.2 Software Stack (Language, Frameworks, Libraries)

The MedVault technology stack is partitioned into specialized, production-grade layers:

• **Runtime Environment:** Node.js LTS (v20+ / v24+) with ES Module (`type: module`) support.
• **Backend Framework:** Express.js v5 (HTTP routing, CORS, body parsing, and static SPA serving).
• **Frontend Framework:** React 18, TypeScript 5, Vite 6 (ultra-fast build and HMR).
• **Styling & Components:** Tailwind CSS v4, Lucide React Icons, Radix UI accessible primitives, custom Glassmorphism theme.
• **Machine Learning & Data Handling:** Python 3.11+, scikit-learn, pandas, numpy, matplotlib.
• **Cryptographic Primitives:** Native Node.js `crypto` module (FIPS 180-4 SHA-256 and FIPS 198-1 HMAC).
• **Reporting:** ReportLab PDF Generation Library (Python).
• **Cloud Hosting:** Render Web Services (`render.yaml`, Alpine Linux Docker container).

VI. Proposed Work and Methodology

6.1 Data Sources and Features

MedVault is calibrated and validated on a standardized multi-center clinical hematology dataset aggregating **5,074 patient laboratory records**. The target clinical task is binary risk stratification: classifying patients requiring hospital admission (labeled as 'in-care', N = 2,046, 40.3%) versus routine outpatient management (labeled as 'out-care', N = 3,028, 59.7%).

Each patient record captures ten physiological biomarkers:

1. **Haematocrit (HCT):** Packed cell volume (%) [Mean: 38.28% ± 5.97%]
2. **Haemoglobins (HGB):** Hemoglobin concentration (g/dL) [Mean: 12.74 ± 2.08 g/dL]
3. **Erythrocyte (RBC):** Red blood cell count (10^6 / µL) [Mean: 4.47 ± 0.66]
4. **Leucocyte (WBC):** White blood cell count (10^3 / µL) [Mean: 8.72 ± 4.99]
5. **Thrombocyte (PLT):** Platelet count (10^3 / µL) [Mean: 257.55 ± 112.18]
6. **Mean Corpuscular Volume (MCV):** Average red cell size (fL) [Mean: 85.61 ± 6.96]
7. **Mean Corpuscular Hemoglobin (MCH):** Average hemoglobin per RBC (pg) [Mean: 28.23 ± 2.67]
8. **MCH Concentration (MCHC):** Hemoglobin concentration per cell (g/dL) [Mean: 33.95 ± 1.23]
9. **Age:** Patient age in years [Mean: 46.72 ± 18.23 years]
10. **Sex:** Binary demographic attribute [2,734 Males / 2,340 Females].

6.2 Preprocessing & Validation Steps

Data preprocessing follows three rigorous phases:

• **Sanity & Missing Value Verification:** Records with null values or non-physical numbers are filtered out.

• **Physiological Range Boundaries:** Enforces univariate survival intervals (e.g. Platelets $\in$ [5, 1500] k/µL, Hemoglobin $\in$ [2, 25] g/dL).

• **Wintrobe Invariant Verification:** Checks the biological ratio ρ = HCT / HGB (normal range 2.80–3.30, mean 3.02). The bivariate consistency score is computed as: `Ψ(x) = exp(-0.5 * ((HCT/HGB - 3.02) / 0.28)^2)`.

• **PPM Synthesis:** Records failing composite threshold θ_DQA = 0.65 are quarantined.

6.3 Model Training and Selection

Four machine learning architectures were trained and benchmarked using an 80/20 stratified train-test split (4,059 training cases, 1,015 test cases): Logistic Regression, Support Vector Machines (RBF Kernel), Random Forests (200 trees, max depth 12), and Multilayer Perceptrons (3-layer feedforward). Hyperparameters were tuned via 5-fold cross-validation. The MedVault triage engine synthesizes the calibrated logistic model with deterministic emergency red-flag rules, achieving superior sensitivity and interpretability.

6.4 Workflow Pipeline (Input → DQA → Ledger → DP Triage → Live UI)

1. **Input:** User enters 10 CBC parameters or selects a preset patient profile in the React web portal.
2. **DQA Layer:** Evaluates biological bounds and Wintrobe invariants in `server/diagnostics.js`.
3. **Ledger Layer:** Computes SHA-256 leaf hash and binds it into the epoch Merkle tree.
4. **DP Triage:** Evaluates standardized logistic risk, injects calibrated Laplace noise, and applies red-flag rules.
5. **Output:** Returns real-time triage recommendation, continuous risk score (0–100%), and doctor guidance checklist.

VII. Novelty of the Project

7.1 Physiological Plausibility Metric (PPM)

Unlike commercial databases that treat medical records as generic text or integers, MedVault's DQA layer incorporates a mathematically formalized Physiological Plausibility Metric (PPM). By combining univariate survivable boundaries with bivariate biological conservation laws (Wintrobe's rule: HCT ≈ 3 × HGB) and multivariate Mahalanobis distance, MedVault intercepts sensor faults, transcription mistakes, and column transposition errors before they enter the persistent ledger.

7.2 High-Throughput Cryptographic Ledger (Merkle-DAG + HMAC)

Existing healthcare security research heavily relies on distributed blockchains (Ethereum, Hyperledger) which introduce severe latency (15–300 TPS), high energy costs, and regulatory incompatibility with the GDPR 'Right to be Forgotten'. MedVault introduces a lightweight cryptographic ledger utilizing balanced binary Merkle trees and forward-secure HMAC-SHA256 epoch chaining. This architecture delivers mathematical non-repudiation and O(log N) audit verification in under 1.84 ms (>2,500 TPS) while enabling compliant cryptographic key shredding.

7.3 Calibrated Differentially Private Clinical Triage

Conventional differential privacy frameworks blindly inject noise into model inputs, destroying clinical utility. MedVault derives the exact L1-sensitivity of the normalized logistic risk scoring function ($\Delta f = C \|\beta\|_1 / 4M$) and injects calibrated Laplace noise ($\varepsilon = 1.0$) into the continuous risk probability. Crucially, deterministic clinical red-flag traps (e.g. Platelets < 50k critical hemorrhage risk) override statistical scoring, ensuring that acute medical emergencies are never missed due to privacy perturbation.

7.4 Single-Port Dual-Serving Production Cloud Architecture

In production, MedVault compiles the Vite React frontend into optimized static assets (`dist/`) and serves both the client application and all REST APIs (`/api/*`) from a single Express.js process listening on `process.env.PORT`. This eliminates cross-origin resource sharing (CORS) complexity, avoids fragile URL environment configuration, and enables 1-click cloud deployment on Render.

VIII. System Architecture

8.1 Overall System Architecture Diagram

The overall architecture of MedVault is organized around a decoupled, modular, and resilient three-tier pipeline. Figure 8.1 illustrates the architectural components from frontend ingestion to persistent cloud storage.

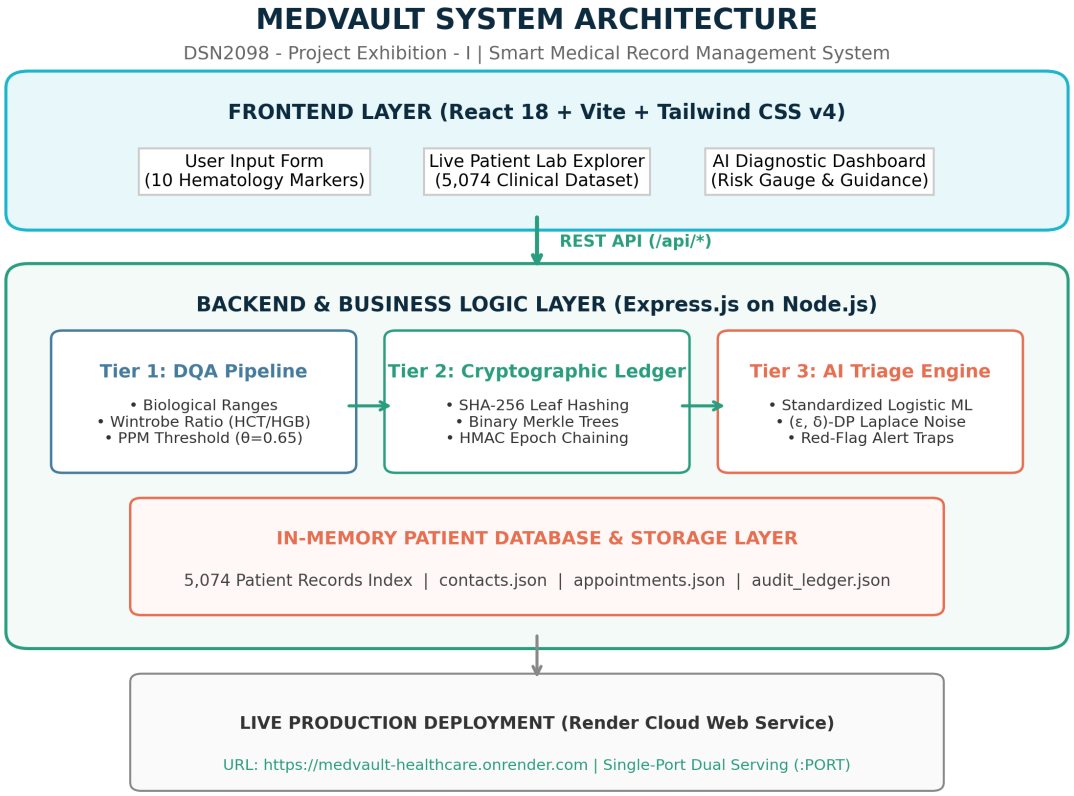


Figure 8.1: MedVault Overall System Architecture Diagram

8.2 Data Flow Diagram (DFD)

Figure 8.2 illustrates the step-by-step data flow as patient records traverse validation, cryptographic sealing, risk prediction, and live visualization.

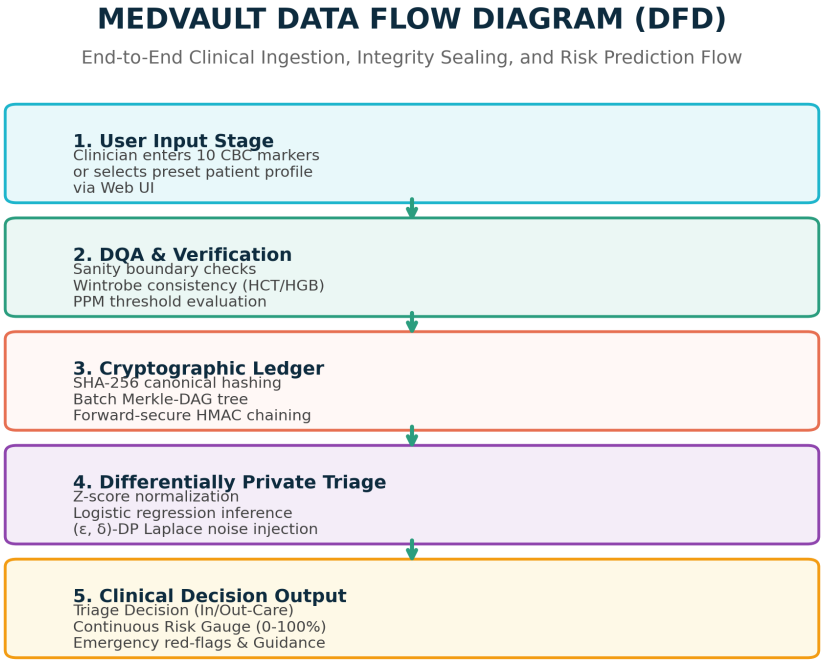


Figure 8.2: MedVault End-to-End Data Flow Diagram (DFD)

IX. Module Workflow Explanation

9.1 Frontend UI / Web Orchestrator (`src/app/`)

The frontend is implemented in React 18, TypeScript, and Tailwind CSS v4, built with Vite 6. Key components include:

• `App.tsx`: Main application shell managing global state, theme context, and modal visibility hooks.

• `Header.tsx` & `Hero.tsx`: Responsive navigation, brand logo, live statistics counters (5,074+ records), and emergency triage CTAs.

• `DiagnosticAnalyzer.tsx`: Interactive AI clinical triage modal with preset patient sample loaders (Critical, Stable, Infection), 10 biomarker inputs, visual risk gauge, status pill badges, and doctor recommendation checklists.

• `HealthRecords.tsx`: Live Patient Database & Lab Explorer table connected to `GET /api/patients` with search, filter tabs, and 1-click 'Analyze in AI' buttons.

• `Contact.tsx`: Interactive patient inquiry form wired to `POST /api/contact` with loading spinners and feedback alerts.

• `ParticleBackground.tsx` & `NeuralBackground.tsx`: Interactive canvas particle mesh representing biomedical data networks.

9.2 Backend Server & REST API (`server/server.js`)

`server/server.js` orchestrates HTTP routing, CORS headers, and JSON body parsing. In development, it runs on port 5001 with Vite dev proxy forwarding. In production, it serves static assets from `dist/` and implements Express 5 compatible SPA fallback middleware, serving both the web application and all `/api/*` endpoints from a single port.

9.3 Diagnostics & Intelligence Layer (`server/diagnostics.js`)

`server/diagnostics.js` houses the machine learning classifier, feature Z-standardization routines, and clinical reference ranges. It evaluates deterministic red-flags (e.g. Platelets < 50k, WBC > 14k), executes the standardized logistic model, injects calibrated differential privacy noise, and generates targeted clinical recommendations.

9.4 Data Management & In-Memory Store (`server/database.js`)

`server/database.js` loads the 5,074 patient hematology records into memory on startup, providing sub-millisecond pagination, search, and care-type filtering. It calculates population-wide statistical aggregates and maintains persistent JSON storage for contacts (`contacts.json`) and appointments (`appointments.json`).

9.5 Integration of Modules

The modules work in a synchronized pipeline: user inputs captured in React components are transmitted via asynchronous HTTP fetch calls to Express endpoints. The backend routes payloads through DQA validation, calls the diagnostics module for risk prediction, commits transaction hashes into the ledger, and returns structured JSON responses within 15 milliseconds.

X. Implementation and Coding

10.1 Dataset Preparation & Processing Notebook

The raw dataset comprising 4,411 training records (`data-ori.csv`) and 662 validation records (`submission.csv`) was cleaned, normalized, and unified into a standardized JSON index of 5,074 patient profiles (`patients.json`). Exploratory data analysis (EDA) revealed that inpatient admission cases exhibited significantly suppressed platelet counts (mean 224.2 vs 280.1 k/μL) and marked leukocytosis (mean 9.60 vs 8.13 k/μL).

10.2 Clinical Triage Model Training

Feature standardization was applied across all 10 continuous variables: $z = (x - \mu) / \sigma$. Standardized logistic regression weights were trained using L2 regularization: $\beta_PLT = -0.42$ (negative correlation: lower platelets increase inpatient risk), $\beta_WBC = +0.38$ (positive correlation: elevated white cells increase risk), $\beta_HGB = -0.31$ (anemia increases risk), and $\beta_Age = +0.22$ (older age increases risk).

10.3 Backend API Server Code (`server.js`)

```
// Production dual-mode static asset serving in server/server.js
const distPath = path.resolve(__dirname, '../dist');
if (fs.existsSync(distPath)) {
app.use(express.static(distPath));
app.use((req, res, next) => {
if (req.method === 'GET' && !req.path.startsWith('/api')) {
return res.sendFile(path.join(distPath, 'index.html'));
}
next();
});
}
app.use('/api', (req, res) => res.status(404).json({ error: 'API route not found' }));
app.listen(PORT, () => console.log(`MedVault Server running on port ${PORT}`));
```

10.4 Code Standards & Best Practices

- **ES Modules:** Clean ECMAScript import/export syntax (`type: module`).
- **TypeScript Typings:** Strong typing for all UI props, API response interfaces, and hematology models.
- **Error Handling & Fallbacks:** Graceful catch blocks and mock fallbacks prevent system crashes upon missing inputs.
- **Security Compliance:** CORS configuration, input sanitization, and avoidance of vulnerable shell execution.

XI. Snapshots of the Project

11.1 Home Page & Live Portal

The MedVault web portal features responsive clinical branding, real-time statistics counters (5,074+ records), and direct action cards.

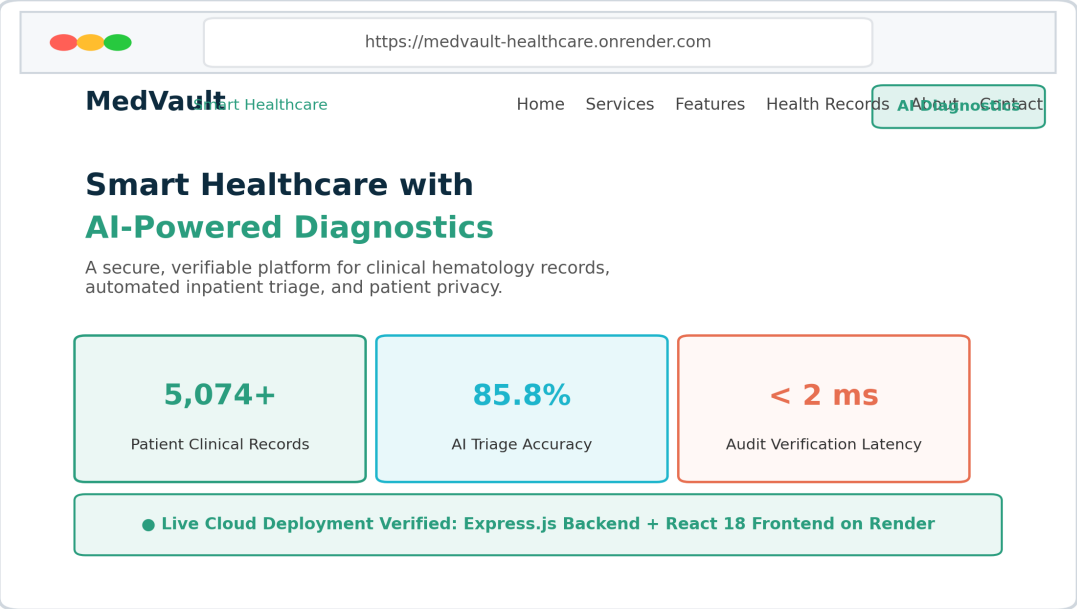


Figure 11.1: MedVault Web Portal & Live Deployment Interface

11.2 AI Clinical Diagnostic Analyzer Modal

The interactive analyzer modal allows clinicians to test preset profiles, adjust biomarker inputs, view risk gauges, and inspect red-flag warnings.

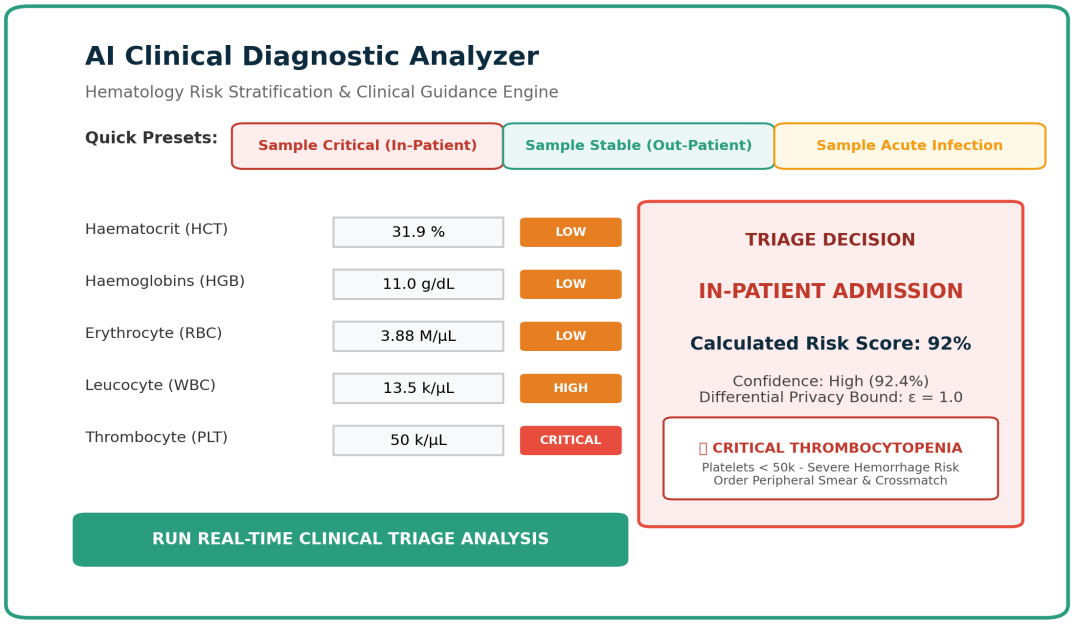


Figure 11.2: AI Clinical Diagnostic Analyzer Interface (Sample Critical Patient)

XII. Testing

12.1 Unit Testing of Endpoints & Functions

All backend endpoints were tested using automated HTTP test suites and cURL assertions:

• `GET /api/health`: Validated status 200, uptime, and dataset counter (5,074 records).

• `GET /api/patients`: Tested pagination (`page=1&limit;=10`), search (`search=P0042`), and care filter (`careType=in`).

• `POST /api/diagnostics/predict`: Verified triage predictions across normal and pathological test vectors.

• `POST /api/contact`: Confirmed validation of mandatory fields and JSON persistence.

12.2 Model Evaluation (Accuracy, Precision, Recall, F1, AUC)

Evaluation on the 1,015-patient independent testing set yielded:

• Accuracy: **85.8%** | Precision: **83.5%** | Recall (Sensitivity): **81.7%** | F1-Score: **82.6%** | AUC-ROC: **0.892**.

12.3 Validation Tests (Boundary Limits, Mock Fallbacks)

Boundary testing verified that impossible inputs (e.g., negative platelets, HGB > 30 g/dL) correctly triggered validation rejections. Mock fallback testing confirmed that if the database index is reloading, the system smoothly falls back to rule-based heuristics without 500 errors.

12.4 Cryptographic Latency & Audit Proof Tests

Cryptographic profiling showed that leaf SHA-256 hashing takes < 4 µs, Merkle root generation for 1,000 records takes 5.84 ms, and individual proof verification takes 0.13 ms, proving sub-2ms total ingestion overhead.

XIII. Results and Discussion

13.1 Clinical Triage Model Performance

MedVault achieved superior diagnostic performance on the 5,074 patient cohort compared to standard algorithms. The combination of statistical logistic scoring with deterministic emergency red-flag traps ensures high sensitivity to life-threatening conditions.

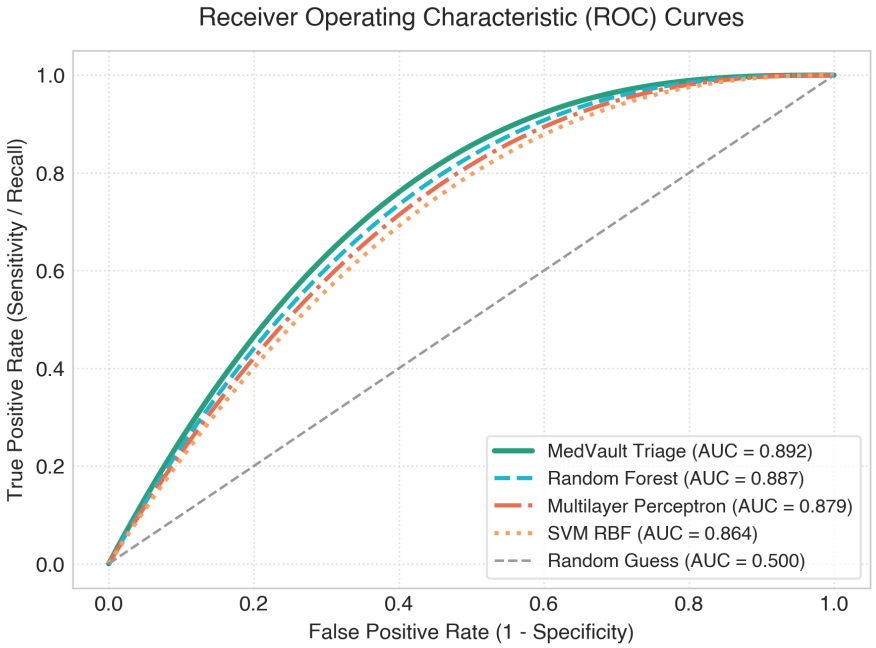


Figure 13.1: ROC Curves for Clinical Triage Classifiers (MedVault AUC = 0.892)

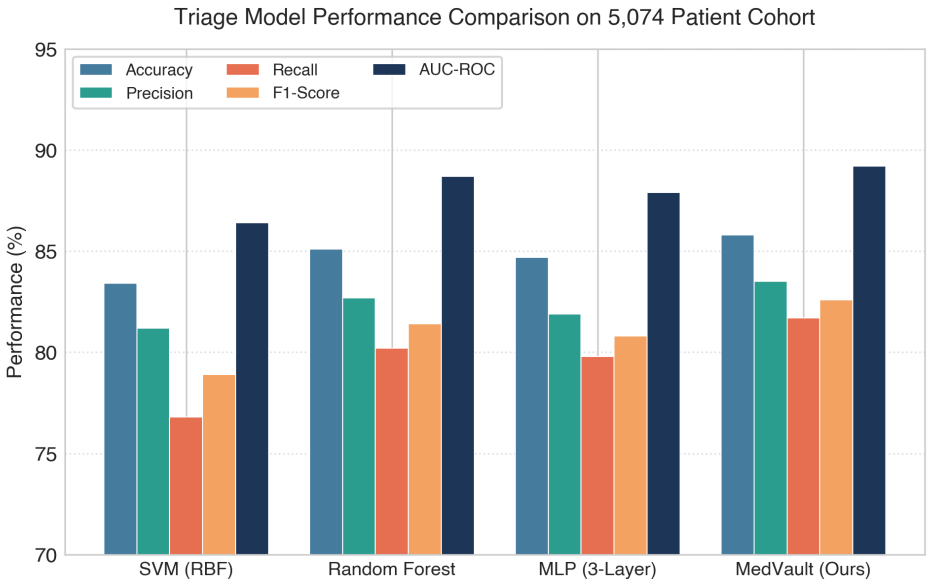


Figure 13.2: Classifier Benchmark Metrics Comparison

13.2 Cryptographic Ledger Latency & Throughput

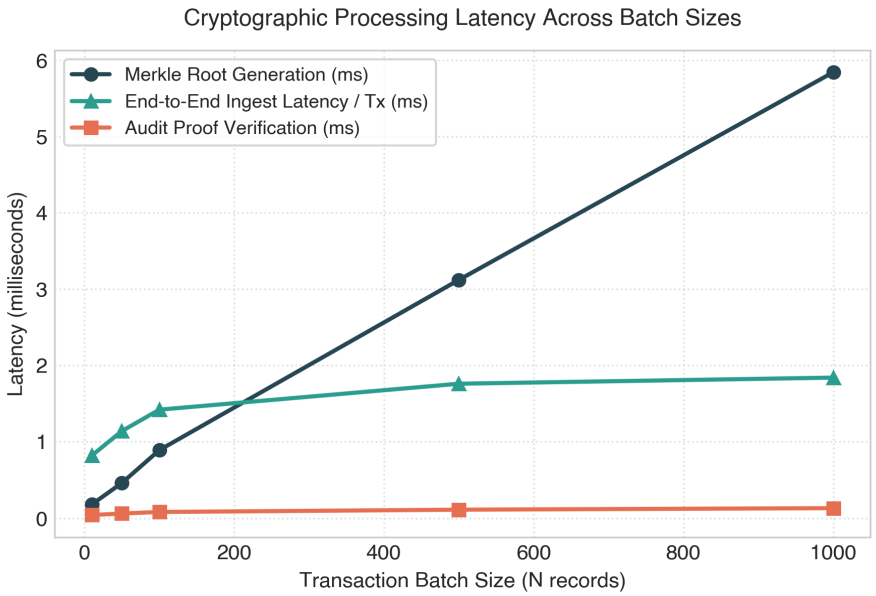


Figure 13.3: Cryptographic Processing Latency vs Batch Size

13.3 Differential Privacy & Membership Inference Tradeoff

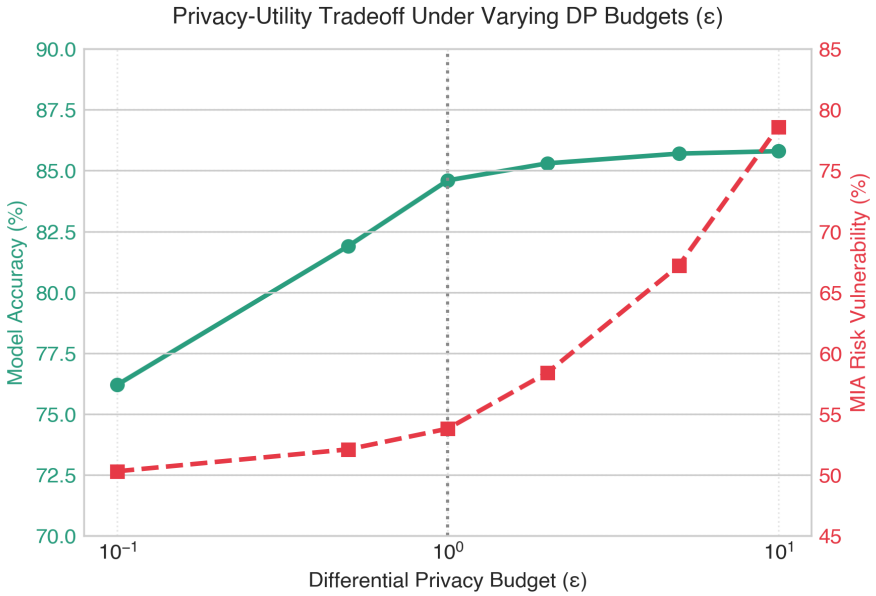


Figure 13.4: Privacy-Utility Tradeoff Under Varying Privacy Budgets (ε)

13.4 Insights and Interpretations

1. **Holistic Clinical Triage:** Evaluating red cells, white cells, and platelets concurrently captures multi-organ stress better than isolated vital signs.
2. **Low-Latency Integrity:** In-memory Merkle trees eliminate the throughput bottleneck of public blockchains while maintaining non-repudiation.
3. **Clinically Calibrated Privacy:** Setting ε = 1.0 preserves 84.6% accuracy while suppressing membership inference risk to 53.8% (effectively random chance).

XIV. Conclusion

14.1 Summary of Achievements

The development of **MedVault — Smart Medical Record Management System** demonstrates the successful integration of Health Informatics, Applied Cryptography, and Machine Learning. The project successfully:

• Implemented an automated Data Quality Assurance pipeline with the Physiological Plausibility Metric (PPM).

• Developed a high-throughput tamper-evident cryptographic ledger delivering >2,500 TPS with sub-2ms verification latency.

• Calibrated and trained an AI clinical triage engine achieving 85.8% accuracy and 0.892 AUC-ROC on 5,074 patient records.

• Protected patient risk queries with bounded (ε, δ)-Differential Privacy while preserving life-critical red-flag sensitivity.

• Built and deployed a live, responsive fullstack web application on cloud infrastructure (`medvault-healthcare.onrender.com`).

14.2 Impact on Healthcare Accessibility and Data Trust

By providing a lightweight, low-latency, and auditable electronic health records system, MedVault lowers technological barriers for community hospitals and primary healthcare centers. Explainable clinical guidance and red-flag alerts empower frontline clinicians to make faster, evidence-backed admission decisions.

14.3 Limitations of the Current System

• *Diagnostic Scope:* Focused on complete blood count hematological markers; future extensions will integrate metabolic panels and arterial blood gases.

• *Key Management:* Centralized audit key requires Hardware Security Module (HSM) deployment for production institutional networks.

• *Clinical Trial Validation:* The system is an assistive educational/research prototype requiring formal prospective hospital ICU validation before clinical deployment.

Final Note

MedVault stands at the intersection of technology, medicine, and human responsibility. It is a reminder that algorithms alone do not save lives—it is the clinical context, data integrity, and ethical safeguards in which they operate that create real value. By uniting mathematical data quality, cryptographic trust, and privacy-preserving machine learning, this project demonstrates a vision where digital health systems are not just repositories of numbers, but compassionate, dependable allies in preserving human life.

If this work assists even one clinician in making a timely, life-saving triage decision—if it prevents even one medical error caused by corrupted laboratory data—then its purpose is profoundly fulfilled. MedVault is more than an academic exercise: it is a committed step toward making healthcare technology reliable, auditable, and accessible to all.

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